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Chapter 12
Headsup With High Blinds: The Jam-or-Fold Game

In Chapter 11, we discussed three half-street toy poker games. For each of these games, we were able to find the optimal strategies - full solutions to the game. Solving toy games can be useful in terms of real poker because each of these games has a particular lesson that we might attempt to apply to real-life poker games. This is even more valuable because in general, real poker games are too complex to solve directly, even using computers.

In this chapter we will work through a toy game as a preliminary stepping stone, and then discuss and provide the computed solution to a game which is of real and immediate use in no-limit holdem games being played every day. The end product of this analysis is the solution to headsup no-limit holdem ***where the first player must either jam (move all-in) or fold.***

We consider these games here in the midst of our half-street games even though they have a slightly different structure because they are more like half-street games than any other type. When we look at jam-or-fold games, instead of considering ex-showdown value, we will consider the total value of the game. By doing this, we create a situation where the first player to act has similar options to Y in the half-street games. He can fold (analogous to a check), which gives him value zero. Or he can bet the amount stipulated by the game, after which the other player may call or fold. Viewed in this fight, jam-or-fold games are an extension of half-street ideas.

Before we begin discussion of this game, however, we invite the reader to test his intuition by considering the following question, the solution to which will be revealed through this analysis. Some readers who know something about the jam-or-fold game may already know the answer.

Two equally skilled players contest a headsup no-limit holdem freezeout where the button must either jam or fold. Each has sixteen chips to begin the hand. The blinds are one chip on the button and two chips in the big blind. With what percentage of hands should the button, playing optimally, move all-in?

We now consider our first toy game, which is a simple game of jam-or-fold with static hand values based on the [0,1] distribution.

Example 12.1 - [0,1] Jam-or-Fold Game #1

Both players have equal stacks of S units.
Each player receives a number uniformly from [0,1].
The player without the button (call him the ***defender,*** X) posts a blind of 1 unit.
The button (call him the ***attacker,*** Y) posts a blind of 0.5 units and acts first.
The attacker may either raise all-in to S units (***jam),*** or fold, surrendering the small blind.
If the attacker goes all-in, the defender may call or fold.
If there is a showdown, then the hand with the lower number wins the pot.

The first thing to notice about this game is that the attacker has just two alternatives -jamming and folding. When a player has two strategic options, where one of the options involves putting money in the pot and one of them is folding, putting money in the pot with any hand that is worse than a hand you would fold is always dominated. Because of this, the first player's strategy will consist of just two regions - strong hands with which he will jam, and weak hands with which he will fold, hi the same way, the second player will call with his strongest hands and fold his weakest hands. Unlike other situations where we bet our strong hands along with the weakest hands, it does us no good here. In fact, if our jamming frequency is *x*, jamming with the strongest

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x hands dominates all other sets of jamming hands.

Our two players' strategies can be expressed directly as two numbers, the attacker's strategy y and the defender's strategy x. These numbers are thresholds; with hands better than y, the attacker jams, and with hands better than x, the defender calls.

Additionally, we know that y and x for the optimal strategies are ***indifference points.*** This means that when Y (the attacker) holds the exact hand y, he is indifferent to jamming or folding. Likewise, when X (the defender) holds the exact hand x, the defender is indifferent to calling a jam or folding. We also know that X will never call with a hand worse than y, Y's jamming threshold - if he did, he could never win the pot. Using this information, we can solve the game directly using indifference equations, as in the last chapter. For this game, we'll deviate from our usual practice and calculate the total value of the game, including the blinds.

The expectation of folding for Y is simply $-\frac{1}{2}$ units (he loses his small blind).

$\langle Y, \text{fold}\rangle = -\frac{1}{2}$

Since X will never call with a hand worse than y, X always wins when he calls. Then the expectation of jamming with the threshold hand y is:

$\langle Y, \text{jam} \mid y \rangle$ = (defender calls) $(-S)$ + (defender folds) $(+1)$

For a uniform distribution such as the [0,1] distribution, the probability of an interval being selected is equal to the length of that interval. Then the defender calls with probability x and folds with probability $(1 - x)$.

$\langle Y, \text{jam} \mid -y \rangle = (x)(-S) + (1 - x)(+1)$
$\langle Y, \text{jam} \mid y \rangle = -xS + 1 - x$

We know also that at the threshold value, Y will be indifferent between jamming and folding. Hence the value of these two strategic options will be equal. Setting this equal to the value of the folding alternative:

$-xS + 1 - x = -\frac{1}{2}$
$x(1+S) = \frac{3}{2}$

$x = 3/(2 + 2S)$ **(12.1)**

The expectation of folding to a jam is -1 unit, while the expectation of calling a jam with the threshold hand x is:

$\langle X, \text{call} \mid x \rangle$ = (attacker has a better hand)$(-S)$ + (attacker has a worse hand)$(+S)$

The attacker jams a total of y hands (we know that $x < y$ because X will never call with a hand worse than y.) Of these hands, $(y - x)/y$ are worse than the calling threshold x and x/y are better. For example, suppose that Y jams with 50% of hands, while X calls with 30%. ($y = 0.5$, $x = 0.3$) Then $(y - x)/y = 2/5$ of Y's jamming hands would be worse than X's calling threshold, and the remaining $x/y = 3/5$ of Y's jamming hands would be able to beat it.

$\langle X, \text{call} \mid x \rangle = p(\text{Y wins})(-S) + p(\text{X wins})(+S)$
$\langle X, \text{call} \mid x \rangle = (x/y)(-S) + [(y - x)/y](+S)$

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<X, fold> = - l

Setting these equal so that X will be indifferent between calling and folding, we have:

$(S)(y-2x)/y = -1$
$Sy - 2Sx = - y$
$S - S2x/y = -1$

$y = S2x/(S + 1)$ **(12.2)**

Remembering Equation 12.1:

$x = 3/(2 + 2S)$

Substituting for x gives us:

$y = 3S/(1 + S)^2$ **(12.3)**

These two equations, then, are the optimal strategies for this game: the attacker jams with $3S/(1 + S)^2$ of the hands, and the defender calls with $3/(2 + 2S)$ of the hands. We can test these solutions intuitively to see if they make sense. For example, with stacks of 5 units, the attacker jams $(3(5)/(l + 5)^2) = 15/36$ of the time, and the defender calls $(3/(2 + 2(5)) = {}^1/_4$ of the time. With stacks of 10 units, the attacker jams $(3(10)/(1 + 10)^2) = 30/121$ of the time, while the defender calls only $3/(2 + 2(10)) = 3/22$ of the time. We can see that at the extreme, with stacks of, for example, a million, both players' frequencies are close to zero.

Is this game good for the attacker or for the defender? We can determine the value of the game to find out. Here we use the matrix method and look at the matchup possibilities.

X →	[0, x]	[x, y]	[y, 1]
Y			
[y, 1]	-0.5	-0.5	-0.5
[x, y]	-S	1	1
[0, x]	0	1	1

When the attacker folds, the defender gains 0.5 units. This happens 1- y of the time.

When the attacker jams and the defender folds, the attacker gains 1 unit. This happens $y(1 - x)$ of the time.

When the attacker jams and the defender calls, there are two cases. The first case is when the two players are both within the caller's threshold. In this case, they break even, as each will have all numbers from zero to x with equal probability, and so their distributions are equal. The second case is when the attacker is above the defender's threshold. In that case, the defender wins S units. This occurs when the attacker jams with a hand worse than the defenders calling threshold and the defender has a calling hand, which has probability $(y - x)(x)$.

So we have:

$< \Upsilon > = p(\text{Y folds})(-{}^1/_2) + p(\text{Y jams, X folds})(+1) + p(\text{Y jams, X calls and } x > y)(-S) +$
$p(\text{Y jams, X calls and } x < y)(0)$
$< \Upsilon > = - (1 - y)/2 + (y)(1 - x)(1) + (y - x)\, x\, (-S)$

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$$< \Upsilon > = -\frac{1}{2} + y\left[\frac{1}{2} + 1 - x - Sx\right] + Sx^2$$

$$< \Upsilon > = -\frac{1}{2} + y\left[\frac{3}{2} - (1+S)x\right] + S\left(\frac{3}{2(1+S)}\right)^2$$

$$< \Upsilon > = -\frac{1}{2} + y\left[\frac{3}{2} - \frac{3}{2}\right] + \frac{9S}{4(1+S)^2}$$

$$< \Upsilon > = \frac{9S}{4(1+S)^2} - \frac{1}{2}$$

Using this valuation function, we can examine the value of the game as the stack size grows from 1 to infinity. With stack sizes from 1 to 2, the game favors tire attacker, while above 2 units, the game favors the defender. As the stacks go to infinity, the advantage of the defender simply goes to the 0.5 units he wins when the attacker folds, because the attacker must fold so often. So overall, this game is a pretty poor one for the attacker.

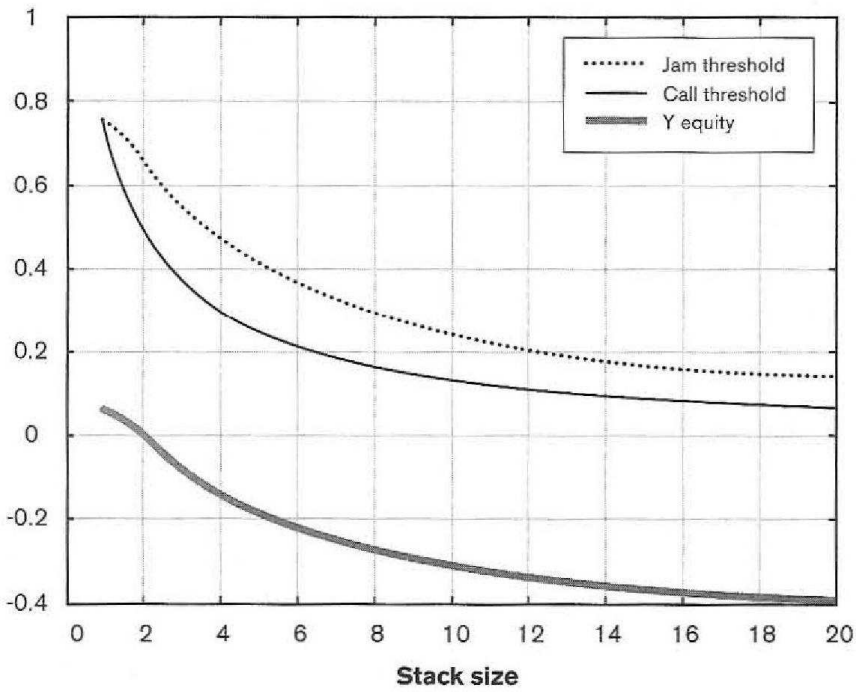

Figure 12.2. **Jam or Fold Game #1**

When we play poker, however, the best hand preflop doesn't always win. In fact, this is dramatically not the case. Even if the defender calls with a fairly strong distribution, the attacker will normally not have less than 33% of equity in the pot. And so our next toy game is a variation on the previous game, but where the worst hand preflop has a 33% chance of winning at showdown.

Example 12.2 - [0,1] Jam-or-Fold Game #2

Both players have equal stacks of S units.
Each player receives a number uniformly from [0,1].
The player without the button (call him the ***defender*** *ox* X) posts a blind of 1 unit.
The button (call him the ***attacker*** *or* Y) posts a blind of 0.5 units and acts first.
The attacker may either raise all-in to *S* units ***(jam),*** or fold, surrendering the small blind.
If the attacker goes all-in, the defender may call or fold.

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If there is a showdown, then the hand with the lower number has $^2/_3$ equity in the pot.

This game is quite similar to the previous games, and we can solve it in much the same way, except for two important differences. In the previous game, when the stacks were very small (for example, at $S = 1$), the attacker still only jammed with 75% of his hands, while the defender only called with 75% of his. This because hands very close to 1 had virtually no value. But in this game, even the very worst hand has $^1/_3$ equity in the pot because of the rules of the game. So at stack size 1 the attacker will have higher equity from putting his chips in than from folding. Secondly, in the previous game the defender could not gain from calling with a hand worse than any hand with which the attacker would jam. However, in this game, this is no longer always the case because the defender always has at least $^1/_3$ of the pot.

We divide the game into two cases.

Case 1: S < 3.
When the stacks are less than 3 units, we can conclude something about the defender's strategy. If the attacker jams, the defender will be getting at least two to one immediately. Since the defender always has at least $^1/_3$ of the pot, he should always call - that is, $x = 1$. This creates a situation where the defender really has no strategic options, as folding is dominated by calling. Thus, the attacker should simply maximize his equity against the defender's static strategy.

<Y, jam> = p(attacker has best) (pot value) (pot equity) + p(attacker has worse) (pot value) (pot equity) - (cost of bet)
<Y, jam> = p(attacker has best)$(2S)(^2/_3)$ + p(attacker has worse) $(2S)$ $(^1/_3)$ - (S)

The defender will call with all hands. Hence, if the attacker jams with y_1, the defender will have a hand better than y_1, with p = y_1, and a worse hand with $p = 1 - y_1$,

p{attacker has best hand) = $1 - y_1$
p(attacker has worst hand) = y_1
<Y, jam | y_1 > = $(1 - y_1)(2S)(^2/_3) + y_1 (2S)(^1/_3) - (S)$
<Y, jam | y_1 > = $S/3 – 2Sy_1/3$
<Y, fold | y_1 > = $- ^1/_2$

Then the attacker should jam whenever:.

$S/3 – 2Sy_1/3 > - ^1/_2$
$2S - 4Sy_1 > -3$
$y_1 < (2S + 3)/4S$

Since he will jam whenever his hand value y_1 is greater than this, his threshold value y is:

$y = (2S + 3)/4S$

The optimal strategies for the game when the stacks are less than three, then, is for the attacker to jam whenever this equation is satisfied and fold otherwise, and for the defender to call all the time. If we examine the play at different stack sizes, we can see that this makes sense. For example, at a stack size of less than 1.5 units, the attacker is now getting two to one to jam even if he will be called l00% of the time. So he should always jam below that threshold. As we can see by substituting 1.5 for S, this satisfies the equation for all hands the attacker can hold.

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Likewise, at the stack size 3, we see that the attacker should jam with $^3/_4$ of his hands. If we calculate his equity from jamming with the hand 0.75, we find that it is - 0.5 units, just the same as if he had folded.

Case 2: S > 3.
At stack sizes greater than 3, however, the defender is no longer getting two to one on his calls. As a result, he cannot call with all hands and do better than folding. In fact, he can no longer call with hands worse than the attacker's jamming threshold. If he does, he will simply have one-third of the equity in the pot, and since the stacks are large enough, this will have worse expectation than folding. As a result, this game is fairly similar to the first jam-or-fold game we examined (Example 12.1), except that the best hand has just $^2/_3$ of the equity in the pot.

We can solve the game using the same indifferences from that game (with the modifications regarding the distribution of the pot). One identity we will use is that the if there is a showdown (i.e., a jam and a call), the player with the best hand wins S/3 units, while the player with the worst hand loses S/3 units.

<Y, jam | y > = (defender calls) (-S/3) + (defender folds) (+1)

The defender calls with probability x and folds with probability (1 - x).

<Y, jam | y> = (x)(-S/3) + (l – x)(+l)

<Y, jam | y> = (-xS/3) + 1 - x **(12.4)**
<Y, fold > = - $^1/_2$

Setting this equal to make Y indifferent between jamming and folding:

$-xS/3 + 1 - x = -\,^1/_2$
$xS + 3x = 9/2$

$x = 9/2(S + 3)$ **(12.5)**

For X, the expectation of folding to a jam is 1 unit, while the expectation of calling a jam with the threshold hand x is:

As in Example 12.1, the attacker jams a total of y hands (we know that $x < y$ from previously). Of these hands, $(y - x)/y$ are worse than the calling threshold x and x/y are better.

<X, call | x>= p(attacker has better) (-S/3) + p((attacker has worse) (+S/3)
<X, call | x> = (- S/3)(x/y) + (+S/3)(y - x)/y
<X, call | x> = (S/3)(y - 2x)/y

<X, fold> = - l

Setting these equal to make X indifferent between folding to a jam and calling it:

$(S/3)(y-2x)/y = -1$
$S(y - 2x) = -3y$
$y = 2xS/(S + 3)$

$y = 9S/(S+3)^2$ **(12.6)**

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So the optimal strategy is for the attacker to jam with $9S/(S+3)^2$ of his hands, and the defender to defend with $9S/(S+3)$ of his hands.

Recall that in Example 12.1, the play was comparatively tight; that is, at stacks around 10, the attacker jammed about a fourth of the time. Contrast that with this game. Here the attacker jams with $(9(10)/(10+3)^2) = 90/169$, or more than half his hands at a stack size of 10! The defender then calls with $(9/(6 + 2(10)) = 9/26$ of his hands, or a little more than a third. This might seem quite loose. After all, can it really be right to jam ten times the blind more than half the time, when one is a 2:1 underdog when called?

t is. The reason this seems intuitively wrong is because we are used to considering situations where we "risk X to win Y," putting in a fixed amount and having our chance of succeeding (by the other player folding or winning a showdown) result in a win of a different fixed amount. By that logic, it does seem rather silly to risk ten to win one if the opponent is going to call a third of the time (roughly). However, this is not the case here. Recall that even when the attacker is called, his loss is simply $S/3$, not S. So in a sense, the attacker is putting in just three and a third to win one and a half (the dead money in the pot).

If we compare the solutions to Game #2 (Example 12.2) with $S > 3$ to the solution to Game #1 (Example 12.1), we can see this relationship more strongly:

Game 1:
$x = 3/(2 + 2S)$
$y = 3S/(1 + S)^2$

Game 2:
$x = 9/(2S + 6)$
$y = 9S/(S + 3)^2$

The solution to game 2 is simply the solution to game 1, but substituting S/3 for S. This is because the ***effective stack size*** of Game 2 is just S/3 - this is the amount that the better hand gains when there is a showdown. We can use this concept of effective stack size to calculate the value of the game, G, by simply substituting S/3 into the game value formula we found in Game #1.

$$\langle \Upsilon \rangle = -\,{}^{1}/_{2} + {}^{3}/_{4}\, S/((S/3+1)^2)$$

$$G = -\frac{1}{2} + \frac{27S}{4(S+3)^2}$$

However, this is only the game value for $S > 3$. There are two other regions of stack size that have different solutions. If $S < {}^{3}/_{2}$ then both players play all hands and the game value is zero. In the region between ${}^{3}/_{2}$ and 3, then we found in Case 1 of the last example that:

$y = \frac{2S+3}{4S}$ and x = 1, so
$\langle \Upsilon \rangle = p(\text{Y folds})(-\,{}^{1}/_{2}) + p(\text{Y jams})p(x_1 > y)(S/3)$

$$\langle \Upsilon \rangle = -\frac{1}{2}(1-y) + y(1-y)\frac{S}{3}$$

$$\langle \Upsilon \rangle = (1-y)\left[-\frac{1}{2} + y\frac{S}{3}\right]$$

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$$< \Upsilon > = \frac{2S-3}{4S}\left[-\frac{1}{2}+\frac{2S+3}{12}\right]$$
$$< \Upsilon > = \frac{(2S-3)^2}{48S}$$

Note that the three game value regions are continuous, as we expect, that is:

$< \Upsilon >$ = 0, for $S \leq {}^3/_2$
$< \Upsilon > = (2S-3)^2 /48S$, for ${}^3/_2 \leq S \leq 3$
$< \Upsilon > = - {}^1/_2 + (27S)/(4(S+3)^2)$, for $3 \leq S$

However, if we look at the threshold values *x* and *y* we notice something unusual. That is, for S < 3, *x* = 1, but suddenly when *S* becomes a little larger than 3, *x* = 3/4. That seems unusual. However, it shows that strategies are not continuous with respect to stack sizes or pot sizes.

These two toy games can help us understand how the fact that the best hand does not always win the pot affects play in jam-or-fold games. When the best hand did win the whole pot, proper play was relatively right and the defender had a significant advantage for most stacks. However, when the best hand only won ${}^2/_3$ of the time in Game #2 (Example 12.2), proper play became much looser and perhaps counter to our intuition, the attacker had an advantage all the way up to a stack size of six.

As we shall see in the next section, play in no-lmit holdem is similarly loose and aggressive.

Example 12.3 - Jam-or-Fold No-limit Holdem

Both players have equal stacks of *S* units.
Each player receives a holdem hand.
The player without the button (call him the ***defender)*** posts a blind of 1 unit.
The button (call him the ***attacker)*** posts a blind of 0.5 units and acts first.
The attacker may either raise all-in to 5units (***jam),*** or fold, surrendering the small blind.
If the attacker goes all-in, the defender may call or fold.
If there is a showdown, then a holdem board is dealt and the best poker hand wins.

Here we have what might be considered a logical extension to the jam-or-fold games we have considered previously. However, holdem hands do not conform very nicely to the [0,1] structure; there are no hands which have nearly zero preflop value, and just two that have even 80% against a random hand. Also, hand values are not transitive; this can be easily shown by the famous proposition bet where the shark offers a choice from among AKo, 22, and JTs, provided that he gets to choose once the sucker has chosen. We can, however, analyze this game directly by brute force. Before we do that, however, we think it interesting to examine the situations at the extremes (when the stacks are very large, and when the stacks are very small).

In the following discussion, we will make reference to a technique called fictitious play. We have already made informal use of this technique and it is of exceptional value in solving game theory problems with computers. Much has been written in academia about this topic, but we will summarize the basic idea here. Essentially, if we have a game and a "nemesis" strategy for the game (which given an opponent's strategy returns the maximally exploitive strategy), we can use these to find the optimal strategy.

The process is fairly straightforward. We begin with a candidate strategy for each player; this can be anything, although in most cases if we have a reasonable guess at strategy for both sides,

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the process converges more quickly. Suppose that we have two players, A and B. We take a candidate strategy for both players. We then calculate the maximally exploitive strategy for B given that A plays his candidate strategy. We combine this maximally exploitively strategy with the initial candidate strategy by mixing the two together using a "mixing weight" constant. This mixing weight is some sequence that converges to zero but has enough terms chat strategies can move as much as they need to, such as the harmonic sequence $1/n$.

The mixing process works like this. Suppose that m is the current mixing weight. For each possible decision point, we combine the old strategy S_{old} and the new maximally exploitive strategy S_{new}.

If we apply this process of fictitious play iteratively, it has been proven mathematically that our strategies will converge on the optimal strategy given enough iterations. Often "enough" is some achievable number, which makes this a very useful method for solving games with computers.

Very Large S

When the stacks are very, very large, both sides only play aces. The attacker cannot profitably jam any other hands because the defender can simply pick him off by calling with aces only. However, as we reduce the stack size incrementally, the attacker must add additional hands to his jamming distribution or miss out on additional blind-stealing equity. It should be clear that if the attacker does not add any additional hands, there is no incentive for the defender to call with anything but aces, since the defender is taking down 1.5 unit pots each time the attacker does not hold aces ($^{220}/_{221}$ of the time). Since calling with any non-AA hands when the attacker is jamming with aces loses a great deal, the defender is content to play this strategy until the attacker modifies his.

Above a certain stack size, however, the attacker simply cannot make money by jamming with additional hands, as he loses too much even when he's only called by Aces. You might think that the next hand to be added (once the stacks become a little smaller) would be KK, as this hand is the second-best hand; however, it turns out that this is not the case because at these high stacks, ***card removal*** is more important. Hands that contain aces, even though they do substantially worse against AA than do other pairs, reduce the probability that the other player holds aces by half and thus have higher equity than KK. The hand that performs the very best in jam-or-fold against an opponent who will call with only aces is ATs. We find this by simply evaluating every hand's equity against this strategy.

If the attacker jammed with ATs (the first hand to be added, as we shall soon see) at a stack size of 2000, his equity would be:

<ATs, jam 2000> = p(defender folds) (pot) + p(defender calls)p(attacker wins pot) (pot size) - cost of jam

<ATs, jam2000> = ($^{1222}/_{1225}$) 1.5 + ($^{3}/_{1225}$) ((0.13336) (4000) - 2000)

<ATs, jam 2000> = - 2.09525

for a loss of approximately 2 units per jam. We can use this formula, however, to find the exact stack size, x, at which ATs becomes profitable to jam against a defender who will only call with aces:

<ATs, jam x > = 1.5($^{1222}/_{1225}$) + ($^{3}/_{1225}$) ((0.13336) (x) – x)

Setting this to zero:

1.5($^{1222}/_{1225}$) + ($^{3}/_{1225}$) ((0.13336) (x) – x) = 0

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$x = 833.25$

So at a stack size of 833.25, the attacker can begin jamming with an additional hand, ATs. For stack sizes greater than this, only aces are profitable jamming hands. At the exact stack size above, ATs is actually indifferent to jamming or folding; the attacker can utilize any mixed strategy he chooses with ATs.

Now let us consider the defender's options, once the attacker begins to jam ATs as well as AA. Can he call with any additional hands? The hand (other than AA) that performs best against the distribution {AA, ATs} is actually AKs, with 41.442% equity. If the defender calls with AKs, he actually loses money against this distribution of raising hands. Once the stack size of 833.25 is reached, then the attacker's strategy goes from {jam 100% of AA} to {jam 100% of AA and 100% of ATs}. There is no gradual ramping-up where the attacker jams with just 1% of his ATs; instead, at the moment it becomes profitable to do so, he jams with every ATs.

This interesting phenomenon is analogous to phase transitions in physics. At a critical temperature, the substance suddenly transforms from solid to liquid, the substance has a drastically different configuration for a minute move in the temperature. That is because the equilibrium state may greatly change even for a small perturbation in the system. This can be seen even in the simplest systems, even in one dimension by perturbing f(x) a little there are critical points where the minimum "jumps" from one place to another. In the same way, we can think of the relationship between the size of the pot and the stacks in this game as a temperature - as we raise this ratio, the game becomes warmer. Playing with blinds of 0.5 and 1 and stacks of 800 is a very low-temperature game. Of course, blinds of zero (where it would be co-optimal to fold all hands) would be analogous to absolute zero temperature - where there is no motion whatsoever.

Raising the temperature of the game just a little, we lower the stacks to approximately 833.12. This brings us to the next hand that becomes profitable to jam, which is A5s. (A5s has 13.331% equity against aces). When ATs became profitable, we simply added it full-force to the attacker's jamming range. However, as we add hands, sooner or later the defender will find it profitable to call with another hand.

For the defender, the hand which has the best equity against the range {AA, ATs, A5s} is AKs - and it is actually the favorite if the attacker jams with all three of those hands (50.9%). So if the attacker were to jam with AA, ATs and A5s, the defender could respond by calling with AKs. and the attacker would lose money. If the attacker were to simply exclude A5s from his range, however, his opponent could respond by calling with only AA and the attacker would give up equity. The attacker must figure out a mix of hands with which he can raise such that be maximizes his profit against the best response strategy his opponent can play. Tb do this, he will seek to make the defender indifferent to calling with AKs.

We'll assume the stacks are exactly 833, a value that is just a little warmer than the threshold of 833.12 at which A5s becomes profitable. We'll also only deal with the *change* in equity (for (fee defender) that occurs between the two defender's strategies.

Attacker's hand	**Equity difference (< AKs,call> = *p(call)p(AKs* wins pot) - cost of call)**
AA	(3/1225)((0.12141)(1666)-833) = -1.5447
ATs	(3/1225)((0.70473)(1666)-833) = 0.8353
A5s	(3/1225)((0.69817)(1666)-833) = 0.8085

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Given this, we can see that jamming with A5s is actually better than jamming with ATs when the opponent will call with both AA and AKs.

In order to make the opponent indifferent, we have to make it such that his equity difference from calling is zero. Hence,

(-1.5447) (AA%) + (0.8353) (ATs %) + (0.8085) (A5s %) = 0

We know that AA% will be 1 because aces have far superior equity to the other hands against any hand, so:

0.8353 ATs% + 0.8085 A5s% = 1.5447

We add hands marginally now as they perform best - so A5s goes to 100% as well:

0.8353 ATs% = 0.70935
ATs% = 87.73%

This would indicate that at a stack size of 833, if the attacker jams with the range {AA-100%, ATs - 87.73%, A5s - 100%}, the defender will be indifferent to calling with AKs, and the attacker's equity will be maximized.

Notice the intriguing effect of this - we previously jammed 100% of ATs; now at a stack size less than half a unit smaller, we jam it just a little over 25% of the time. An exhaustive analysis of this game yields any number of oddities like this - in fact, at some stack sizes, hands that were previously jams for the attacker are eliminated completely and replaced by other hands (which do better against the opponent's new range).

Very Small S

Now that we have scratched the surface of what the solution to this game looks like at very large stack sizes, let us consider what happens at very small ones.

First, consider the defender's position. For a stack size *S*, the defender will call with any hands that have positive equity against the opponent's range. The defender will be facing a bet of size *(S* - 1), and a pot of size *(S* + 1). If his equity in the pot when called is *x*, his equity from calling will be:

$x(2S) - (S-1) > 0$
$x(2S) > S - 1$
$x > (S - 1)/(2S)$

So the defender will call when his equity is greater than $(S - 1)/(2S)$.

At a stack size of 1.1, for example, the defender would call when his equity is greater than $^1/_{22}$. If the attacker jammed with all hands, then the defender would always have a call, as the equity of the worst hand (32o) is 32.3% against all hands.

Let's assume, then, that the defender will call with all hands. The attacker will then jam with all hands whose EV is better by jamming than by folding. For any hand, its EV from jamming is simply the outcome of the all-in confrontation against the range of all hands that the defender is calling against, while folding has an EV of 0.

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<ATT, all > (2*S*) - (*S* - 0.5) > 0
<ATT, all > > (*S* - 0.5)/2*S*

So the attacker will jam with all hands whose equity is greater than (*S* - 0.5)/2*S*. At a stack size of 1.1, then, he will call when his equity is greater than $^{0.6}/_{2.2}$, or $^{3}/_{11}$. All hands have at least this equity, so the attacker would jam with all hands. Since neither player can unilaterally improve by changing his strategy, these strategies of jamming all hands and calling all hands are optimal for a stack size of 1.1

As we decrease the "temperature" of this very hot game, it is the attacker who first runs into hands that have insufficient equity with which to jam.

32o has 32.3% equity, so substituting in the above equity formula:

0.323 > (*S* - 0.5)/(2*S*)
S > 1.412

So when the stacks get to 1.412 units, the attacker can no longer profitably jam with 32o. Contrast this to the case of very large stacks, where in all cases, the attacker jammed with more hands than the defender called with.

We can use the same fictitious play technique to examine what the proper strategies are at different stack sizes. For example, consider the stack size 2. Using our EV formula from above, the defender will call whenever his equity hi the pot is greater than (2 - l)/(4), or $^{1}/_{4}$. Assume temporarily that the defender will call with all hands. The attacker's best counter-strategy then, is to jam with all hands that have more than [2 - 0.5)/4, or $^{3}/_{8}$ equity. This excludes a total of 13 hands: 32o, 42o, 52o, 62o, 72o, 82o, 43o, 53o, 63o, 73o, 83o, 32s, and 42s.

Now we look for the best counter-strategy for the defender. Against a range which excludes the above 13 hands, the hand with the lowest equity is still 32o, with 31.72%. As a result, the defender still defends with all hands. Here, both sides have again maximized their equity against then opponent's strategy and neither side can improve unilaterally, so the strategies are optimal.

This may seem a little peculiar to readers who arc experienced with no-limit. Normally it is Correct (or so the conventional wisdom goes, which we will see in a moment agrees with analysis) to call with fewer hands than one would jam. The reason the opposite is true here at very small stacks is that the additional 0.5 units that the attacker has to expend to play the hand is very significant relative to the size of the stacks. As we increase the size of the stacks, we find that this effect is diminished, and the ratio of optimal attacking hands at a specific stack to defending hands at that stack begins to rise again.

Continuing to lower the temperature of the game, we look at stack size of 3. Now the attacker's equity requirement against a defender who will always defend is (3 - 0.5)/6 = $^{5}/_{12}$. Against a defender who calls all the time, this excludes 34 hands:

32o and 32s
42o-43o and 42s-43s
52o-54o and 52s-54s
62o-65o and 62s-64s
72o-75o and 72s-73s
82o-85o and 82s-83s

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92o-94o

The defender's equity to call is (3 - l)/6, or $^1/_3$ Against the attacker's range, the following hands have insufficient equity to call against the attacker's janmiing range:

32o
42o
52o
62o-63o
72o-73o
82o-83o

Now for the attacker, the defender's range has changed, and so his expected value function changes. Now he will jam whenever the following is true:

(% defender calls) (*p*(ATT wins) (new pot) - cost of call + (% defender folds) (pot) > 0

This equation yields a different range of jamming hands; then the defender reacts to this new range, and so on. What we have described here is a loose variant of the fictitious play algorithm we described earlier. Applying that algorithm to the problem, we can find the optimal strategies at a stack size of 3.

Attacker jams:

{22, A2s, A2, K2s, K2, Q2s, Q2, J2s, J2, T2s, T2, 92s, 95, 84s, 85, 74s, 76, 64s, 54s}

Defender folds:

{32, 42, 52, 63-, 73-, 83-}

In fact, we can solve the jam-or-fold game using the fictitious play technique for any stack size. *The* ***jam-or-fold tables,*** which are the optimal solution to the jam-or-fold game for all stack sizes, are reproduced on the next two pages and in the reference section. The tables contain two numbers for each hand - a "jam" number, which is the stack size S below which the attacker should jam, and a "call" number, which is the stack size S below which the defender should call. It is true that (as we saw) at certain stack sizes hands do appear and disappear (in narrow bands), but this can be safely ignored as a practical matter. The upper threshold for these tables is a stack size of 50. Hands that are labeled JAM or CALL are simply calls at any stack size below 50. The few cases where hands disappear and then reappear at significantly higher stack sizes are noted in the tables.

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No Limit Holdem - Jam or Fold tables

(suited hands to the upper right)

Attacker	A	K	Q	J	T	9	8	7	6				
A	JAM	JAM	JAM	JAM	JAM	JAM	JAM	JAM	JAM	JAM	JAM	JAM	47.5
K	JAM	JAM	JAM	JAM	JAM	JAM	JAM	49.2	36.2	32.2	26.4	19.9	19.3
Q	JAM	JAM	JAM	JAM	JAM	JAM	JAM	30.2	29.4	24.4	16.3	13.5	12.7
J	JAM	JAM	JAM	JAM	JAM	JAM	49.8	32.3	18.6	16.2	13.5	10.6	8.8
T	JAM	JAM	44.7	46	JAM	JAM	JAM	35.6	24.7	11.9	10.5	7.7	6.5
9	44.9	24.2	24.3	29.2	31.9	JAM	JAM	36.1	26.8	14.4	6.9	4.9	3.7
8	42.6	18.6	13	14.1	16.4	20.5	JAM	43.2	30.9	18.8	10.1	2.7	2.5
7	40.8	16.1	10.9	8.5	9.9	10.8	15.6	JAM	35.7	23.8	13.9	2.5	2.1
6	34.8	15.3	9.8	6.5	5.7	5.2	7.1	11.2	JAM	29.3	16.3	*	2
5	36.9	14.4	8.9	6	4.1	3.5	3	2.6	2.4	JAM	23.5	**	2
4	34.5	13.1	8.3	5.4	3.8	2.7	2.3	2.1	2	2.1	JAM	***	1.8
3	31.9	12.5	7.5	5	3.4	2.5	1.9	1.8	1.7	1.8	1.8	JAM	1.7
2	29.2	11.6	7	4.6	3	2.2	1.8	1.6	1.5	1.5	1.4	1.4	JAM

Starred cells indicate hands that have broken strategies:
* 63s = 2.3, 5.1 – 7.1
** 53s = 2.4, 4.1 – 12.9
*** 43s = 2.2, 4.8 – 10.0

(suited hands to the upper right)

Defender	A	K	Q	J	T	9	8	7	6	5	4	3	2
A	CALL	CALL	CALL	CALL	CALL	47.3	40.7	35.6	30.9	30.1	25.6	24.7	23.2
K	CALL	CALL	CALL	44.9	31.9	24.2	17.8	15.3	14.4	13.3	12.2	11.4	10.8
Q	CALL	46	CALL	29.1	24.3	16.2	13	10.6	10	8.9	8.5	7.8	7.2
J	CALL	27.4	19.9	CALL	18.4	13.5	10.7	8.8	7.1	6.9	6.2	5.8	5.6
T	CALL	24.1	15.6	13.2	CALL	11.6	9.3	7.4	6.3	5.2	5.2	4.8	4.5
9	39.7	17.5	11.8	9.9	8.5	CALL	8.3	7	5.8	5	4.3	4.1	3.9
8	34.5	14.1	9.8	7.7	6.7	6.1	CALL	6.5	5.6	4.8	4.1	3.6	3.5
7	29.2	12.5	8	6.4	5.5	5	4.7	CALL	5.4	4.8	4.1	3.6	3.3
6	21.9	11.1	7.4	5.4	4.7	4.2	4.1	4	CALL	4.9	4.3	3.8	3.3
5	20.9	10.3	6.8	5.1	4	3.7	3.6	3.6	3.7	42.6	4.6	4	3.6
4	15.6	9.2	6.3	4.8	3.8	3.3	3.2	3.2	3.3	3.5	31.9	3.8	3.4
3	16.8	8.5	5.9	4.5	3.6	3.1	2.9	2.9	3	3.1	3	22.4	3.3
2	16	8.3	5.6	4.2	3.5	3	2.8	2.6	2.7	2.8	2.7	2.6	15.1

For attacker, the number indicated is the stack size (in big blinds) below which he should jam.
For defender, the number indicated is the stack size (in big blinds) below which he should call a jam.
For both players, the stacks indicated are the stack size in play before the blinds of 1/2 and 1 are posted.

Practical Applications and Additional Comments

Most situations that occur in practice do not meet the conditions of these examples - that is, we would not play jam-or-fold with 500 times the blind. But we can see that at extremely large stack sizes, it is optimal to play very few hands, and at extremely small stack sizes, it is optimal to play most, if not all, hands. As the size of the stacks decreases through the range between these extremes, it turns out to be the case that the attacker and defender strategies do in fact add hands at a more or less steady pace.

We can compare the results of jam-or-fold to the results from Jam-or-Fold Game #2 (Example 12.2), where the weaker hand had $^1/_3$ equity in the pot. In that game, at stack sizes of 10, the attacker jammed with $^{90}/_{169}$ (53.2%) of his hands, and the defender called with $^9/_{26}$ (34.6%) of his. In jam-or-fold no-limit holdem, at stack sizes of 10, the attacker jams with $^{774}/_{1326}$ (58.3%)

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of his hands and the defender calls with $^{494}/_{1326}$ (37.3%) of his As you can see, these results are relatively close - the slightly more aggressive strategies in the real no-limit game come from the fact that many matchups are even closer than two to one.

It is of particular interest what types of hands are included in the jamming strategy. In most cases, the hands with the most showdown value against a random hand are included at the upper range of hands (that is, for even very large stack sizes, jamming with high pocket pairs, for example, is indicated). But as the stack size decreases through the typical ranges, a number of hands get added that actually have significantly less showdown equity against random hands. For example, 97s is a jam for stack sizes up to approximately 36 units, despite the fact that it is an underdog against a random hand. The reason for this is that for the attacker, it is irrelevant what equity a jamming hand has against a random hand; what matters is only its equity against the range of hands with which the opponent will call. And middle suited connectors do quite well (relatively speaking) against the types of hands that make up a significant percentage of calling hands, such as high pairs and big aces. The attacker tables also highlight the importance of suited cards - the additional 2.5% that suitedness adds makes it far more attractive to jam with suited cards than their unsuited brethren.

The reader may then ask, "What use is this? I do not play in games where the only options are jamming or folding." It is our contention that playing jam-or-fold for small stack sizes is either the optimal or at minimum a close approximation of the optimal strategy for heads-up no-limit holdem for chose stack sizes and blinds. The proliferation of online single-table tournaments, satellites, and other structures where heads-up high-blind situations are common makes this particularly relevant to the serious player. We concern ourselves, then, with the important question, “When can jam-or-fold reasonably be drought to apply?”

We begin with the following assumptions:

- On an information-hiding basis, it is desirable to play all hands in a consistent manner; that is, raising all hands to a small amount, all-in, calling, or folding.
- Because the distribution of hands that the attacker will play is stronger than the distribution of hands that the defender holds, the attacker should raise in preference to calling.

Call the strategy of playing jam-or-fold *J*. The defender maximizes his expectation against *J* by playing *J* in response. Now consider an alternate strategy J_1, where the first player raises some set of hands (to be determined later) to 2 units.

Let us assume that the defender plays the following (definitely sub-optimal) strategy J_2 in response to the raise to 2 units:

Jam with all hands that would have called a jam.
Fold all hands that would have folded to a jam.

Against J_2, J_1 seems to be stronger than *J;* the attacker only risks 1.5 units and gets the same number of folds as he would by jamming h is entire stack. If the opponent jams in response, he may be able to fold some of his hands and gain equity. At very small stacks, however, this is impossible. For example, at a stack size of 4, the opponent will jam to 4 units (after a raise to 2) with 115 of 169 different hand types, or approximately $^{838}/_{1326}$ (63%) of his hands (this is actually modified for card removal effects based on the attacker's hand, but we ignore this for now).
If the defender jams with 63% of his hands, the attacker clearly has higher equity from calling

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than folding with any hand. Even 32o has more than 30% equity against the range of jamming hands for the defender. So the attacker raising any hand is tantamount to jamming that hand. Since raising to two units in this case is equivalent to jamming, he should only raise the hands in *J*. But this gives the opponent a strategic option (calling and seeing a flop). Since strategic options have non-negative value, the attacker should simply play jam-or-fold at a stack of 4.

Equally obviously, this is not true, at stacks of, say, a hundred. If the opponent played J_2 with such a large stack, this would be excellent for the attacker. He would simply raise to two units with all hands and fold to the defender's jam without a strong hand. Somewhere between these two extremes, however, there is some stack size at which the attacker can likely start to raise smaller amounts profitably. Below this stack size, we can reasonably say that jam-or-fold is essentially the optimal strategy. We're not going to attempt to prove this, or find some specific exact stack size over which this is optimal. Instead, we're looking for a practical recommendation that shows approximately where we can start playing jam-or-fold. We're looking for a stack size above which we can begin to raise a smaller amount profitably with hands outside the jam-or-fold range.

In order to find this stack size, we can look for a situation where for a hand just below the attacker's jamming cutoff, the equity from raising to 2 units and then folding to a jam from J_2 is equivalent to the equity from jamming. At a stack size of eight, the defender will reraise all-in, playing J_2 with $^{602}/_{1326}$, or 45.4% of his hands. Approximately the worst hand that is jammed by the attacker is Q4o. So Q3o is a hand close to the threshold. Against the defender's range of calling hands, Q3o has about 34.8% equity.

The equity of jamming eight units with Q3o is:

<jam, Q3> = *p*(defender folds) (defender's blind) + *p*(defender calls) *p*(attacker wins) (defender's stack) + *p*(defender calls) (*p*(defender wins) (loss of attacker's stack)
<jam, Q3>= (0.546) (+1) + (0.454) (0.348) (+8) + (0.454) (0.652) (-8)
<jam, Q3> = - 0.558 units

This value is close to what we would expect, as it is just below the cutoff for jamming (compared to the -0.5 units for folding). The difference is caused by card removal effects that impact this particular hand.

The equity of raising to two units with Q3o is (again ignoring card removal effects):

<raise to 2, Q3> = *p*(defender rejams) (loss of raise) + *p*(defender folds) (gain of blind)
<raise to 2, Q3> = (0.454) (-2) + (0.546) (+1)
<raise to 2, Q3> = - 0.638

At a stack size of eight, then, we begin to approximate the situation we're looking for; where the attacker is indifferent to jamming *S* units or raising to two units and folding to a jam with hands weaker than his set of jamming hands. As a first-order approximation, then, we could use eight units as the threshold to begin playing jam-or-fold. However, there is another important effect that the above analysis neglects. The defender is not restricted to playing J_1, in fact, it would be foolish for him to do so!

Since the defender can simply call our 2-unit raise, getting 3-1 on his call, the equity of raising two units is decreased. As a result of this, we add a couple additional units to the stack size threshold at which we begin to play jam-or-fold. Therefore, we believe it is likely optimal to play jam-or-fold in the headsup no-limit case at a stack size of 10 units or below, and potentially as

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high as 12 or 13 units when the attacker will be out of position postflop. In situations where the blinds are not of the 0.5-1 structure (such as when the blinds are 2 or 3 chips) or when there are antes (such as when the full table has anted and the play has become headsup by the entire field folding), we simply allocate the antes into the blinds. Two-thirds of the money in the pot is normalized to 1 unit for the purpose of utilizing the jam-or-fold tables. This is only an approximation, but it is useful in practice. Also note that when playing jam-or-fold, only the smaller of the two stacks needs to be below the 10 blind threshold - it doesn't matter how big your stack is if the opponent's stack is small, or vice versa.

Returning to the question we posed at the beginning of this chapter:

Two equally skilled players contest a headsup jam-or-fold no-limit holdem freezeout. Each has sixteen chips to begin the hand. The blinds are one chip on the button and two chips in the big blind. With what percentage of hands should the button move all-in optimally?

Utilizing the jam-or-fold tables, we can see that the set of jamming hands is approximately 61.7% of all hands. That number startles many people on first sight. After all, who would intuitively think that you would put in sixteen chips to win three with more than three-fifths of your hands? This is a powerful idea of which many players are not aware.

These strategies are of great value in the headsup portion of tournaments and in playing in blind vs. blind situations when the blinds are relatively large. These situations occur frequently in, for example, online single-table tournaments where the blinds escalate quickly. In addition, the jam-or-fold game shows the great value of jamming preflop in no-limit holdem. By playing jam-or-fold even at high stack sizes, a weaker player can fairly effectively neutralize a strong player's postflop skill, giving up only a modest amount of equity by doing so.

Key Concepts

- In jam-or-fold where the best hand wins the entire pot, optimal strategy is fairly tight and play favors the defender for most stacks.
- When the worst hand has significant showdown equity, however, optimal strategy loosens up a significant amount.
- In jam-or-fold no-limit holdem, strategies are significantly looser than most players think.
- When selecting hands with which to jam, the showdown equity against the hands that will call is of paramount importance.
- Jam-or-fold is at least near-optimal for stack sizes up to about 10 or 11 blinds.

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